

- 3 (a) (i) 1 deg C corresponds to $(3840 - 190) / 100 \, \Omega$ C1
 for resistance $2300 \, \Omega$, temperature is $100 \times (2300 - 3840) / (190 - 3840)$
 temperature is 42°C A1 [2]
- (ii) *either* $286 \, \text{K} \equiv 13^\circ\text{C}$ *or* $42^\circ\text{C} \equiv 315 \, \text{K}$ B1
 thermodynamic scale does not depend on the property of a substance M1
 so change in resistance (of thermistor) with temperature is non-linear A1 [3]
- (b) heat gained by ice in melting = $0.012 \times 3.3 \times 10^5 \, \text{J}$ C1
 = $3960 \, \text{J}$
 heat lost by water = $0.095 \times 4.2 \times 10^3 \times (28 - \theta)$ C1
 $3960 + (0.012 \times 4.2 \times 10^3 \times \theta) = 0.095 \times 4.2 \times 10^3 \times (28 - \theta)$ C1
 $\theta = 16^\circ\text{C}$ A1 [4]
 (answer 18°C – melted ice omitted – allow max 2 marks)
 (use of $(\theta - T)$ then allow max 1 mark)